

SHOBHIT PANDEY

Senior Software Engineer

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SUMMARY

Senior Full Stack Engineer with **7+** years of experience building and scaling B2B products end to end, owning both front-end and back-end delivery from customer-facing web applications to cloud-native data platforms. Strong in **Python**, **Django**, **React**, and JavaScript (JS), with hands-on experience across **AWS**, **Azure**, and **GCP** and a track record of improving performance, reliability, and cost. Known for cross-team collaboration, clear communication skills, and a focus on user experience, and for applying practical innovation informed by industry trends in cloud, MLOps, and observability.

EDUCATION

Indian Institute of Information Technology, Vadodara, India

B.Tech (Bachelor of Technology), Computer Science and Engineering | CGPA: **8.06** | May **2015** - May **2019**

EXPERIENCE

Polybee | Senior Software Engineer | Singapore

Feb 2023 - Present

Product and Performance

- Led end-to-end development and delivery of a **B2B** agricultural analytics dashboard (phenotyping, forecasting, heatmaps) using **Django**, **React**, and Ant Design, achieving **< 3** second page loads (down from **15** to **20** seconds).
- Reduced API p95 latency to **150** to **200** milliseconds (down from over **60** seconds) by optimizing **AWS** RDS PostgreSQL queries with table-level indexing and introducing Nginx API caching.
- Delivered **3** to **5** GB heatmap layers smoothly in the UI using CDN-backed distribution and client-side performance techniques such as pagination and asynchronous loading.
- Designed and deployed containerized frontend and backend services on **AWS** using ECS, ECR, ALB, and Route **53**, with GitHub Actions **CI/CD** to improve release reliability and production operations.
- Implemented Redis-backed Celery async workflows for background processing and long-running tasks in the application layer.

Data Platform and Reliability

- Orchestrated and automated an event-driven processing pipeline in **Prefect** handling about **2** TB per week across about **45** runs per day (average runtime about **90** minutes) to meet a **12**-hour delivery target from data capture to dashboard.
- Architected and Implemented **Azure**-based execution using Service Bus Queues, Functions, Storage, Storage Tables, and VMs for scalable and cost-aware processing.
- Reduced pipeline failure rate from about **40%** to under **5%** by implementing replay-safe, idempotent pipeline stages keyed by dataset capture timestamps and client and site identifiers, plus distributed locks, retries with backoff, and restart-from-failure checkpoints.
- Cut idle virtual machine time by about **20%** by enforcing fail-safe cleanup and forced shutdown on failures, preventing orphaned compute and controlling cloud costs.
- Implemented end-to-end observability across pipeline stages using Elastic and Sentry with operational reporting in Supabase and Metabase, reducing time to identify issues to under **2** minutes.
- Configured **Prefect** flows for scheduling, retries, and operational SLAs, with alerts surfaced via n8n and dashboards in Supabase and Metabase.
- Integrated automated alerts and operational notifications using n8n to speed up response and reduce manual monitoring.
- Improved on-time data delivery from about **60%** to **98%** while sustaining **99.9%** uptime for **3** to **4** enterprise customers across Australia and the US.

Cloud Migration

- Executed an **Azure**-to-**GCP** migration in about **1** month using canary A/B runs and client-by-client cutovers, integrating Pub/Sub, Cloud Storage, Compute Engine, and GKE with no interruption to production data delivery.

Leadership, Security, and Documentation

- Deployed and maintained a self-hosted secrets management workflow (Passbolt) behind Tailscale, and owned RBAC and least-privilege permissions across **AWS**, **Azure**, and **GCP** to improve access governance.
- Partnered with R&D, operations, and leadership stakeholders to translate requirements into scalable delivery for enterprise B2B customers.
- Mentored **multiple** interns and led code reviews, design reviews, and task planning across dashboard, pipeline, automation, and observability workstreams.
- Authored internal technical documentation and maintained API collections and documentation in Postman to improve onboarding and reduce support cycles.

OYO Rooms | Software Development Engineer 2 | Gurugram, India

Jul 2022 - Jan 2023

- Built and operated high-traffic, customer-facing hotel booking microservices, improving availability and scalability through backend and deployment improvements.
- Increased automated unit test coverage by about **30%** for Java services to reduce regressions and improve release confidence.
- Developed services using Java, Spring Boot, and Maven, and supported Kubernetes-based deployments and **Azure** cloud workflows.

Maximl Labs | Senior Software Engineer | Chennai, India

Apr 2021 - Jul 2022

- Owned end-to-end modules for an analytics platform, partnering with stakeholders to translate requirements into production-ready features.
- Transitioned a monolithic application toward microservices to improve scalability and independent deployments.
- Developed a serverless media storage service supporting multiple storage providers and pre-signed URLs for secure uploads and downloads.
- Improved analytics by expanding API capabilities and enabling page injection for richer product instrumentation.
- Improved mobile reliability by introducing offline usage patterns for low-connectivity scenarios.
- Streamlined delivery by implementing CI for multiple microservices to reduce manual release effort.

Maximl Labs | Software Engineer | Chennai, India

May 2019 - Apr 2021

- Built backend services for an analytics platform using **Django** and **Django** REST Framework, with Celery for asynchronous workloads and multilingual support.
- Enhanced Angular applications with route guards, reusable npm libraries, HTTP interceptors, and IndexedDB-based client storage.
- Designed scalable solutions for data extraction, analysis, and indexing, delivering detailed specifications and resolving technical conflicts.

CORE SKILLS

Languages: Python, Java, JavaScript (JS)

Back-End: Django, Django REST Framework, Celery, API design, caching, AWS RDS PostgreSQL, database indexing, performance tuning, caching.

Front-End: React, Ant Design, user experience, performance (pagination, async loading), responsive UI

Cloud and DevOps: AWS (ECS, ECR, ALB, Route 53, CloudFront, S3, EC2), Azure (Service Bus Queues, Functions, Storage, Storage Tables, Virtual Machines), GCP (Pub/Sub, Cloud Storage, Compute Engine, GKE), Docker, Kubernetes, Nginx (API caching), Unicorn, Redis, GitHub Actions

Data and MLOps: Prefect orchestration, event-driven pipelines, idempotency, retries and backoff, distributed locks

Observability and Operations: Logging, metrics, tracing, Elastic, Sentry, Supabase, Metabase, n8n alerting, incident response, SLAs

Security: RBAC, secrets management, least-privilege access, Passbolt (Self Hosted), Tailscale